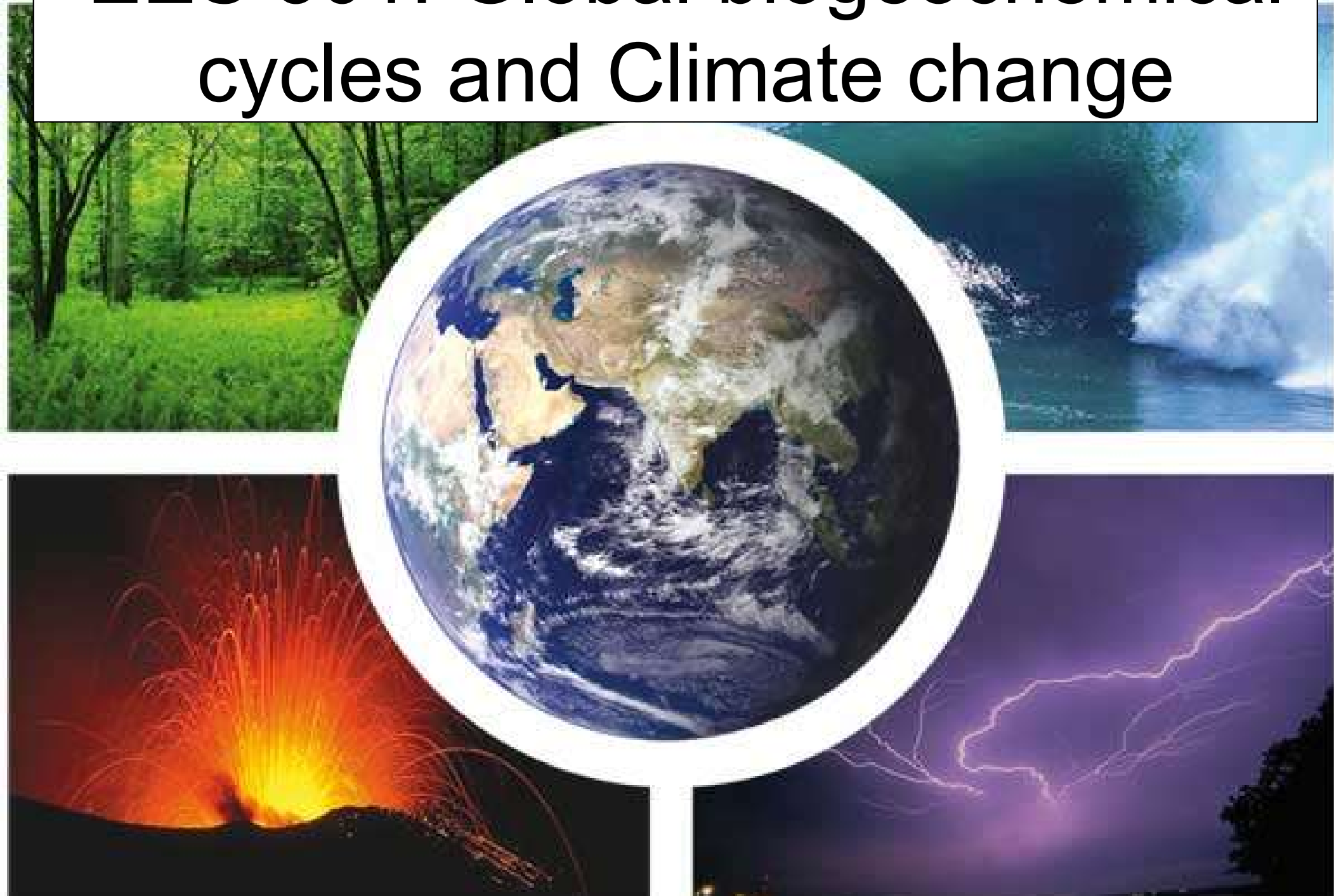


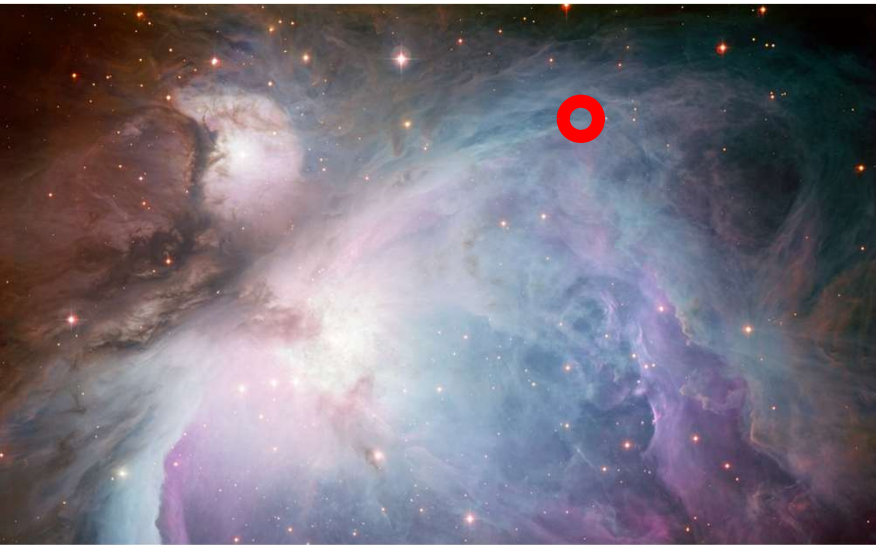
EES 301: Global biogeochemical cycles and Climate change



Origin of solar system – theory must explain why

- 1) Each planet is roughly twice as far as the next inner planet is from the Sun (the Titus-Bode rule).
- 2) 99.9 % of mass is in the Sun; 99 % of angular momentum is in the planets.
- 3) Planets in two groups:
 - *terrestrial (inner)*: Mercury, Venus, Earth, Mars contain heavy elements (O, Si, Fe, Mg) Mercury is mostly Fe ($\rho = 5.4$)
 - *Jovian (outer)*: Jupiter, Saturn, Uranus, Neptune. Jupiter mostly gas and ice ($\rho = 0.7$) Pluto and other trans Neptune bodies ???

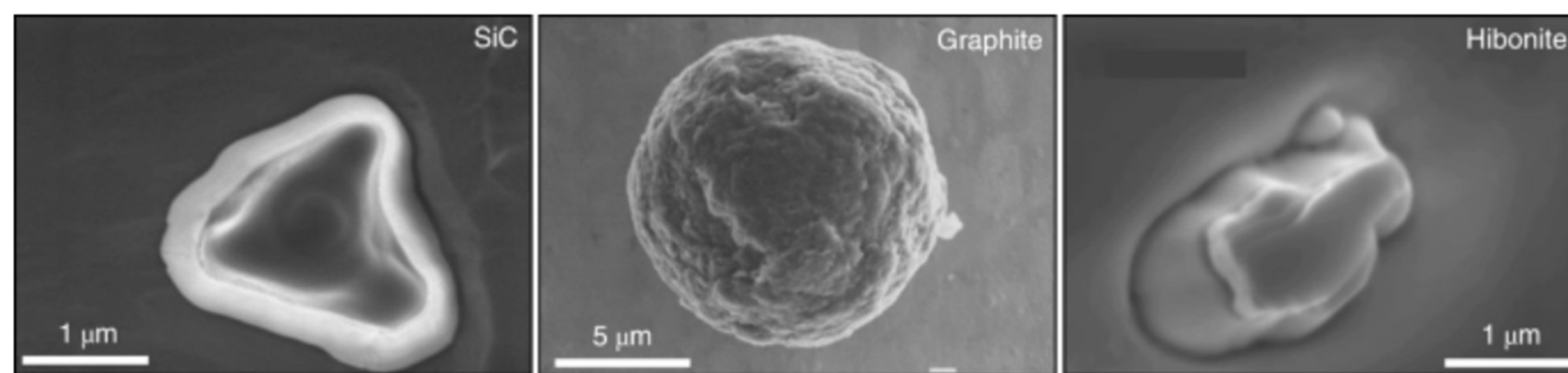
Material in the undisturbed cloud



<https://www.eso.org/public/images/eso1103a/>

Presolar grains

- Only H, He and the noble gases were gaseous. At less than 10 K (Temperature of space) **most condensable material was in the condensed phase.**
- **Presolar grains consisting of diamond (C), graphite (C), silicon carbide (SiC), titanium carbide (TiC), silicon nitride (Si_3N_4), corundum (Al_2O_3), spinel (MgAl_2O_4), hibonite ($(\text{Ca,Ce})(\text{Al,Ti,Mg})_{12}\text{O}_{19}$), titanium oxide (TiO_2), olivine ($(\text{Fe,Mg})_2\text{SiO}_4$) and pyroxene have been found**
- In interstellar space they may have been covered by cross-linked polymers of lower molecular weight substances



The solids of the original cloud can only be studied using presolar grains

Step 2 Supernova shock wave

A shock wave from an exploding supernova in the vicinity injected material from the exploding star into our solar system causing a compression-decompression wave. Due to this the internal gravitational energy exceeded twice the kinetic energy of the cloud and the cloud collapsed in on itself.

$$\frac{3GM^2}{5R} > \frac{3KTM}{\mu m_H}$$

This is supported by the large amount of **nickel 60** (a decay product of short lived unstable isotope **iron 60**). Iron 60 has a “**short**” half-life of **2.6 million years**. The same is true for excess **²⁶Mg**, the decay product of the short-lived radionuclide **²⁶Al** (mean life = 1.03 Myr), **Iron 60** and **Al 26** are only created in significant amounts by nuclear reactions inside certain kinds of stars called asymptotic giant branch (AGB) stars or in a supernovae.

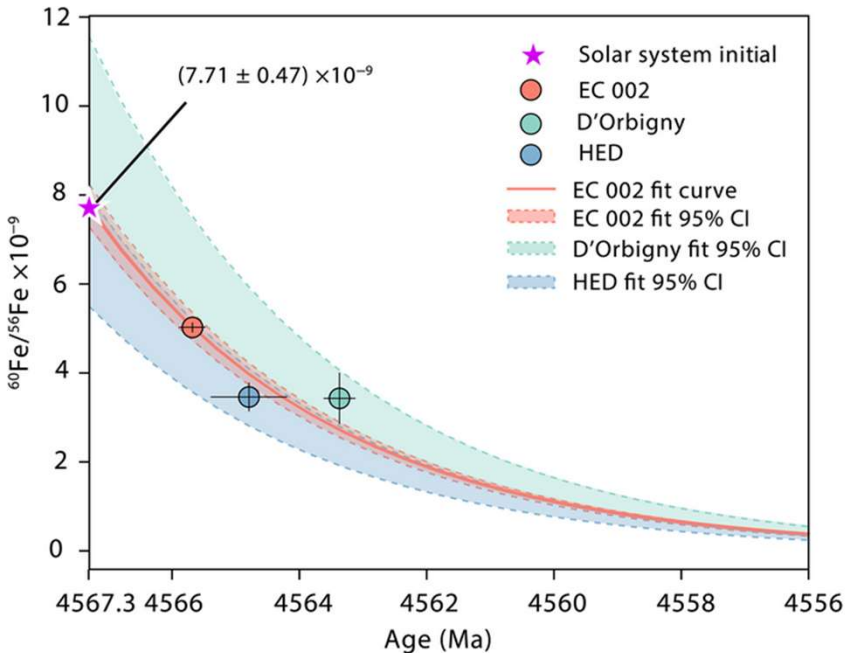


Fig. 2. The solar system initial $^{60}\text{Fe}/^{56}\text{Fe}$ defined by initial $^{60}\text{Fe}/^{56}\text{Fe}$ values and corresponding crystallization ages of EC 002 and other differentiated meteorites.

The $^{60}\text{Fe}/^{56}\text{Fe}$ evolution curve and ranges were constrained by the ages and $^{60}\text{Fe}/^{56}\text{Fe}$ values of EC 002 (red field), D'Orbigny (green field), and HED meteorites (blue field). The crystallization age of angrite D'Orbigny is 4563.37 ± 0.25 Ma dated by the Pb-Pb chronometer (38) and the mantle differentiation time of the parent body of HED meteorites was calculated using their initial $^{53}\text{Mn}/^{55}\text{Mn}$ value [$(4.21 \pm 0.46) \times 10^{-6}$, (40)] relative to that of D'Orbigny [$(3.24 \pm 0.04) \times 10^{-6}$, (39)] and the Pb-Pb absolute age of D'Orbigny. The initial $^{60}\text{Fe}/^{56}\text{Fe}$ values of D'Orbigny and HED meteorites are from (10). All dashed lines show the boundaries of 95% confidence interval (95% CI).

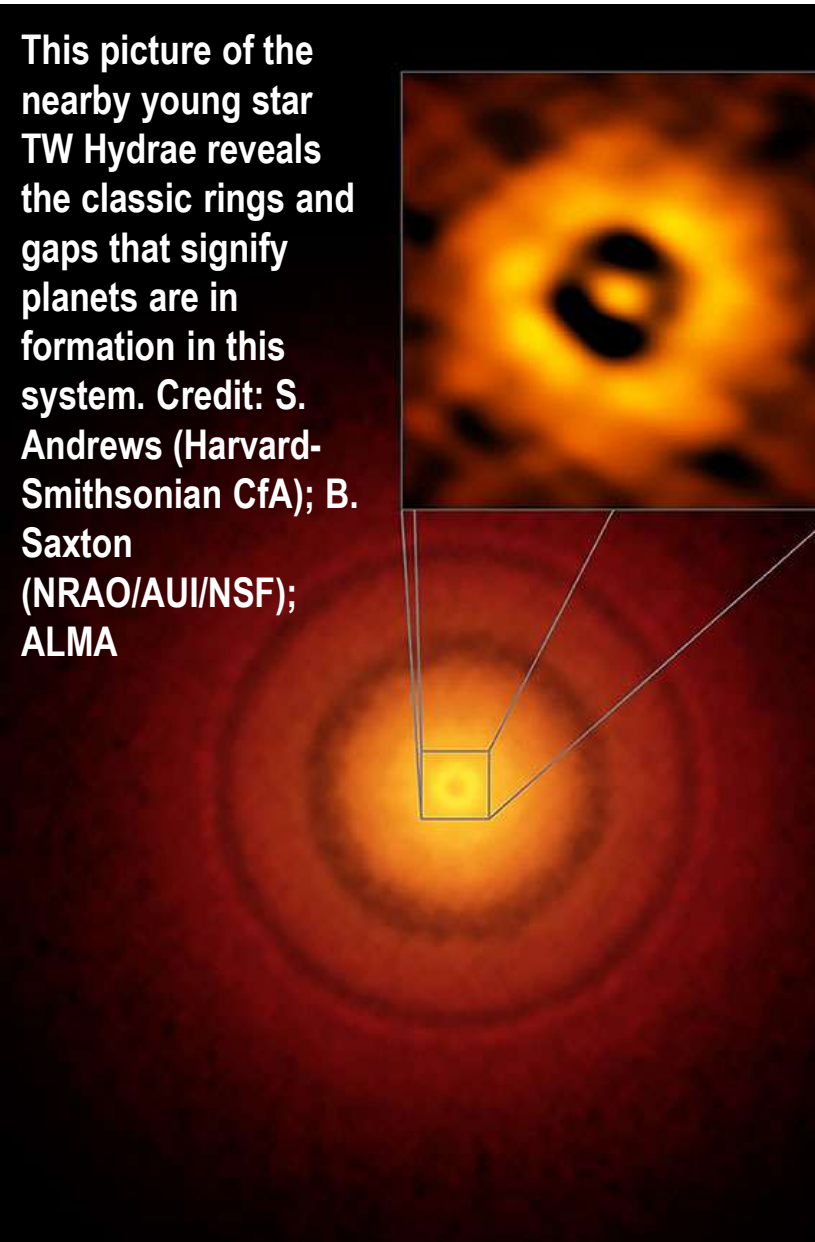
<https://www.science.org/doi/epdf/10.1126/sciadv.adp9381>



Decay products of ^{60}Fe and ^{26}Al are found in all types of early meteorites including differentiated ones. Differentiation happened within < 3 Million years after the supernova.



Step 3: Contraction phase



This picture of the nearby young star TW Hydrae reveals the classic rings and gaps that signify planets are in formation in this system. Credit: S. Andrews (Harvard-Smithsonian CfA); B. Saxton (NRAO/AUI/NSF); ALMA

- During the initial contraction the dimensions of the cloud decreased by a factor of 1000 and the cloud started to spin. The collapse of interstellar material to form a solar nebula takes on the order of 10^5 to 10^6 years depending on the initial conditions. In the case of the solar system the contraction phase was short. In the initial phase of collapse the cloud is transparent to the IR it releases and 10 K cold so condensable material is in the condensed phase.
- However, once nebula densities are reached the cloud becomes opaque to IR and traps the huge amount of potential energy released till in the center hydrogen atoms start to bang into each other so violently that fusion starts and the sun lights up. At this point solar winds start blowing gasses outwards.
- The planets Jupiter and Saturn must have formed due to some version of gravitational collapse early on, as their chemical composition (H, He) matches that of the bulk nebula
- A nebula is hottest and densest near the center near the sun, and cooler and more diffuse with increasing radial distance. In the area where terrestrial planets formed it became so hot that most of the original interstellar grains vaporized and solids had to recondensed from the nebula.

Step 4: Condensation phase



Primitive meteorites called Chondrites preserve an image of how the material in the early solar system looked like. The round Ca–Al-rich inclusions are secondary solids formed during condensation from a >1300 K gas.



Iron rich chondrites called enstatites contain reduced iron and minerals that form at >1400 K

- The calcium–aluminium-rich inclusion or Ca–Al-rich inclusion (CAI) found in carbonaceous chondrite meteorites are probably the oldest substances in the Solar System.
- These are used to date the solar system. The oldest age was measured in an inclusion of the CV3 carbonaceous chondrite Northwest Africa (NWA) 2364 and was dated at 4568.22 ± 0.17 Mya using Pb-Pb dating.
- These are thought to have formed as fine-grained condensates from a high temperature (>1300 K) gas that existed in the protoplanetary disk during the condensation phase
- They survived in small bodies that were not large enough to heat up and differentiate

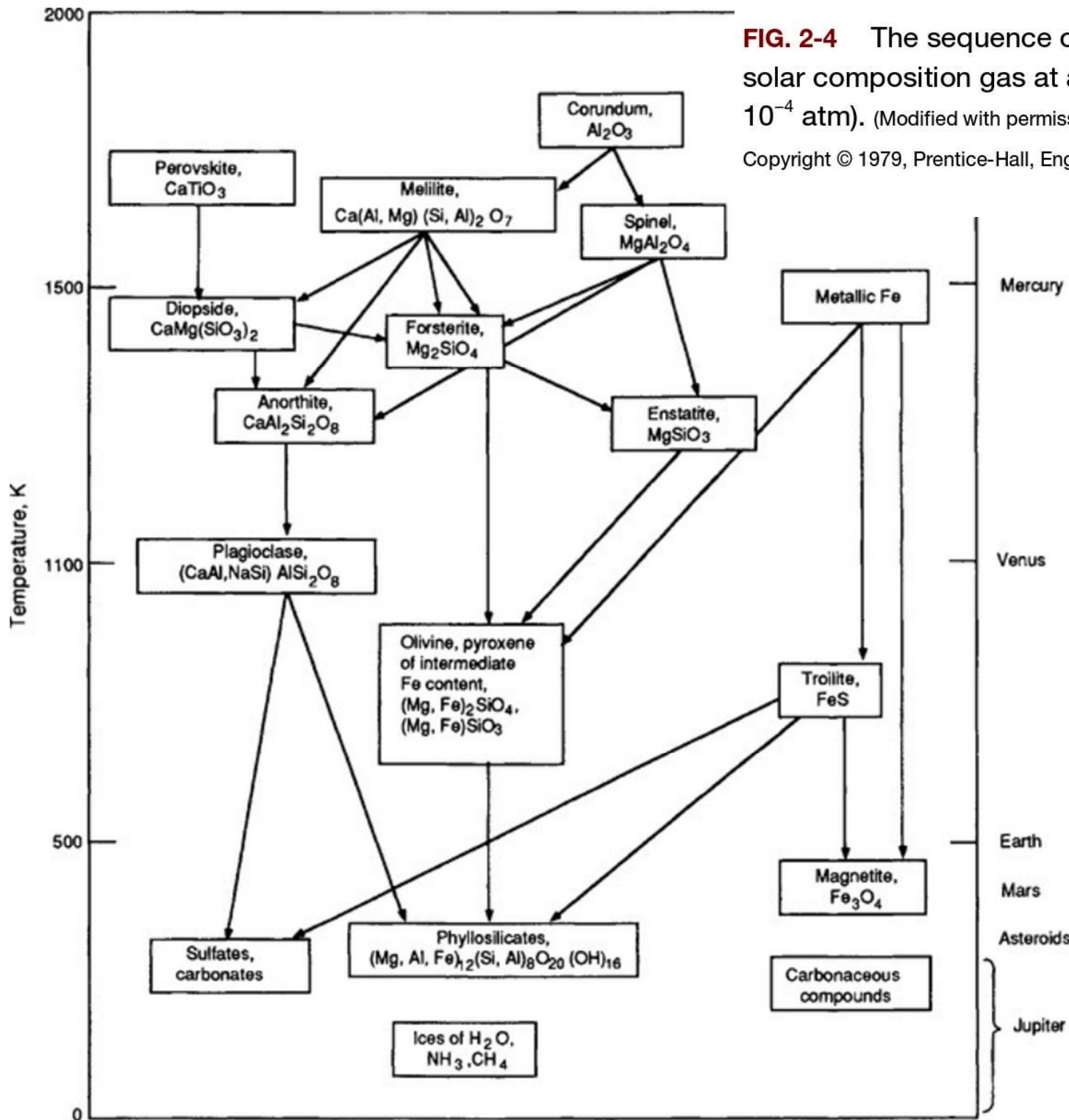
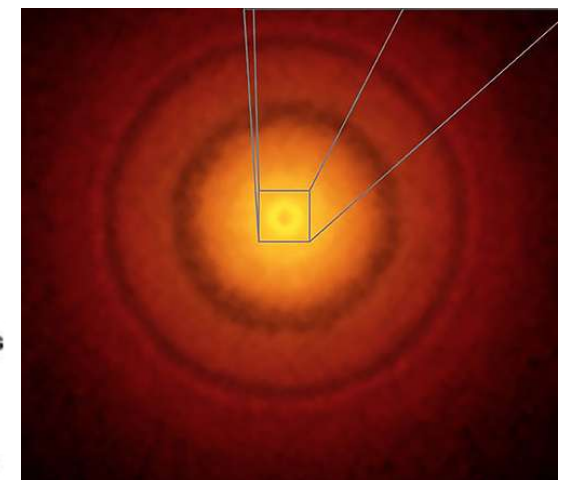
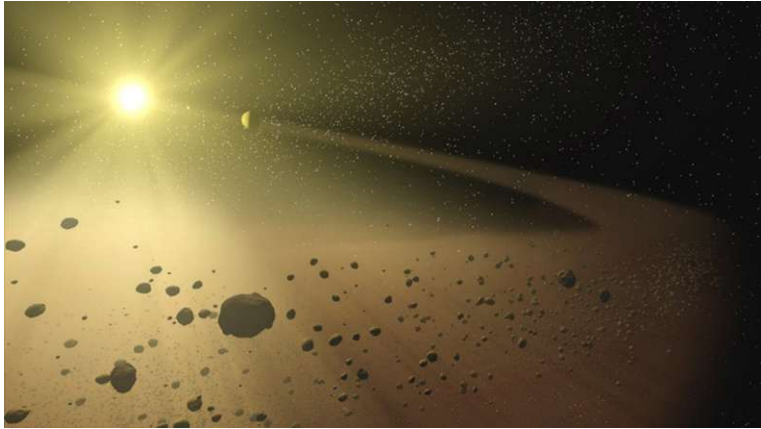


FIG. 2-4 The sequence of condensation of solids from a solar composition gas at a nebular pressure of 10 Pa (ca. 10^{-4} atm). (Modified with permission from J. A. Wood, "The Solar System," p. 162, Copyright © 1979, Prentice-Hall, Englewood Cliffs, NJ.)

The bulk mass of each planet seems to have been formed largely from the material in its feeding zone, an annular ring of the protoplanetary disk that had material condensed at a particular temperature



Step 5: Accretion phase



Picture of the Kuyper belt revealing how the solar system may have looked like in the accretion phase



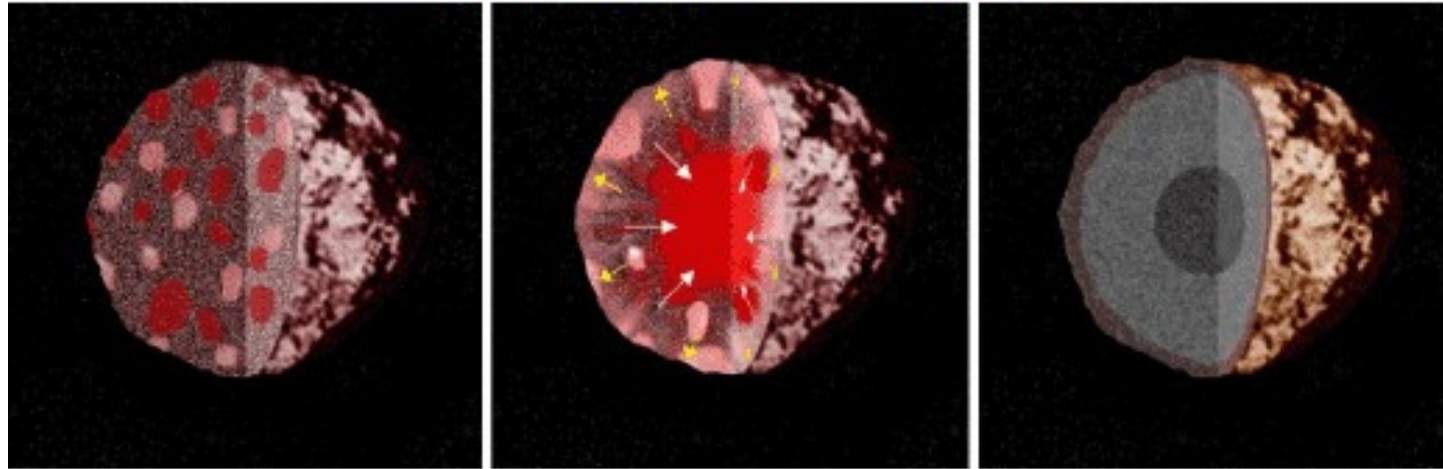
Primitive meteorites called Chondrites preserve an image of how the material in the early solar system looked like

- The enveloping disk of gas and dust collide and forms grains that clump together to form smaller planetesimals.
- **Terrestrial planets grew by multiple collisions and accretion of planetesimals by gravitational attraction mostly from silicate rich and iron rich material.**
- **The outer planets Uranus and Neptun accreted from ice, carbonaceous material and stony material**
- **The terrestrial planets already melted and differentiate in this phase. They lost their initial atmospheres to solar winds till they developed a magnetic field.**
- **At some point in this phase at around 4.51 Ga the differentiated proto-Earth collided with a Mars-sized proto-planet called Theia. This collision tilted Earth's axis and formed the moon.**



Step 6: Planetary differentiation

Differentiation of a Planetesimal



(Frank Granshaw, Artemis Software)

- **Radioactive decay of ^{26}Al and ^{60}Fe produced in the supernova that triggered the formation of the solar system provided a strong heat source that melted all large planetary bodies**
- **Once the melting started heavier metals were drawn towards the centre of gravity and released huge amount of potential energy**
- **This triggered the iron catastrophe (complete melting of the planetary body)**

The planetesimal called Vesta which sent a large number of meteorites to Earth is used to answer questions about planetary differentiation because the body was large enough to melt but small enough to cool fast. The material from Earth and other large planets can't answer these questions because it has been reworked by plate tectonics.



The Vestian asteroids and HED meteorites have been broken from Vesta by a collision of a large object

Within 2–3 million years

Within 4–5 million years

Within 6–7 million years

Accretion completed

Complete or almost complete melting due to radioactive decay of ^{26}Al , leading to separation of the metal core

Progressive crystallization of a convecting molten mantle. Convection stopped when about 80% of the material had crystallized

Extrusion of the remaining molten material to form the crust, either as basaltic lavas in progressive eruptions, or possibly forming a short-lived magma ocean.

The deeper layers of the crust crystallize to form plutonic rocks, whereas older basalts are metamorphosed due to the pressure of newer surface layers.

Slow cooling of the interior

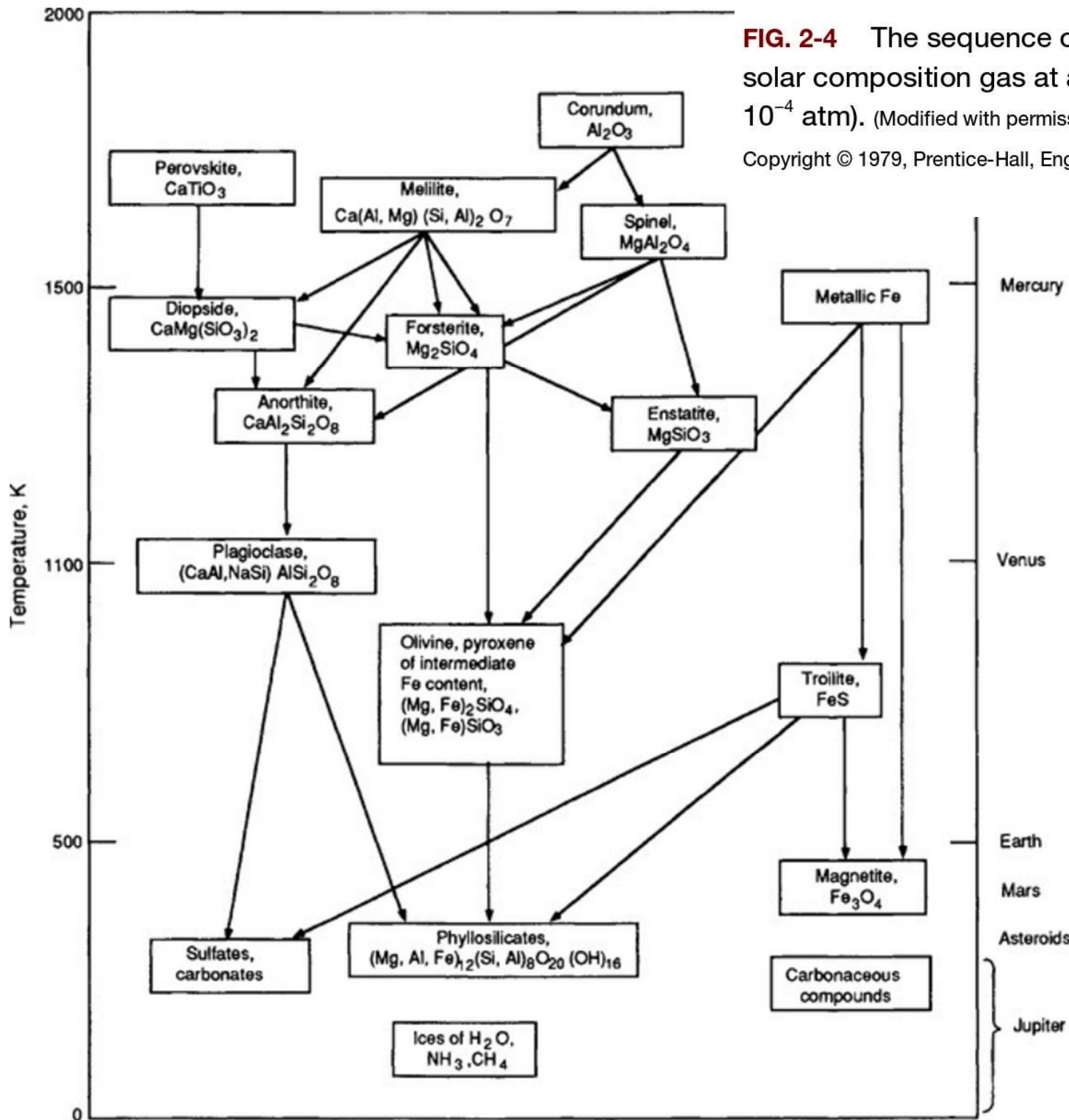


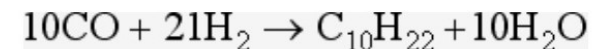
FIG. 2-4 The sequence of condensation of solids from a solar composition gas at a nebular pressure of 10 Pa (ca. 10^{-4} atm). (Modified with permission from J. A. Wood, "The Solar System," p. 162, Copyright © 1979, Prentice-Hall, Englewood Cliffs, NJ.)

The mystery of water and carbon on Earth

Water can easily condense and accrete in the outer part of the solar system beyond the Asteroid belt where methane ice forms as in equilibrium:

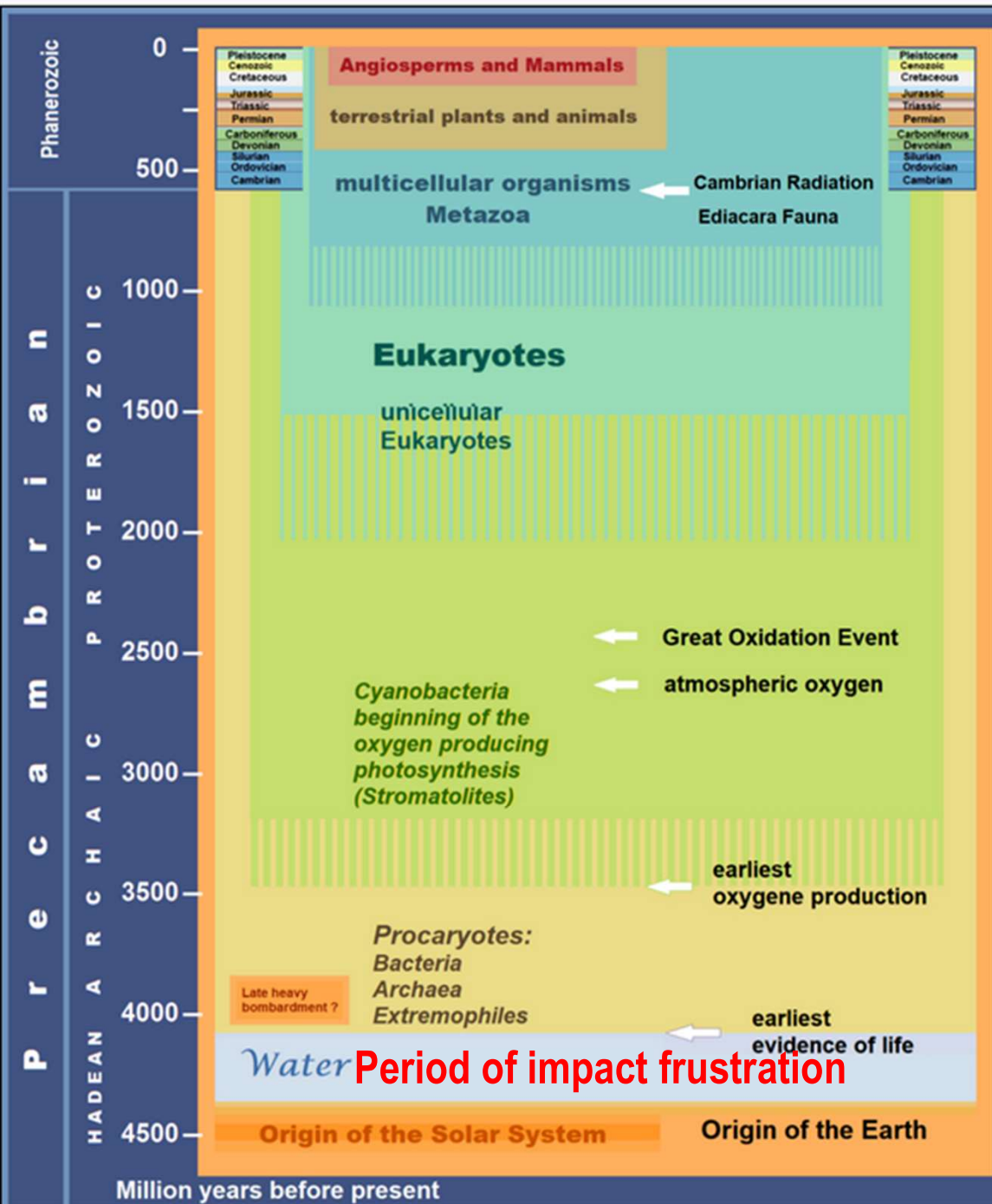


However, equilibrium may have not been reached and some Fisher-Tropsch reaction may have produced solid carbon:

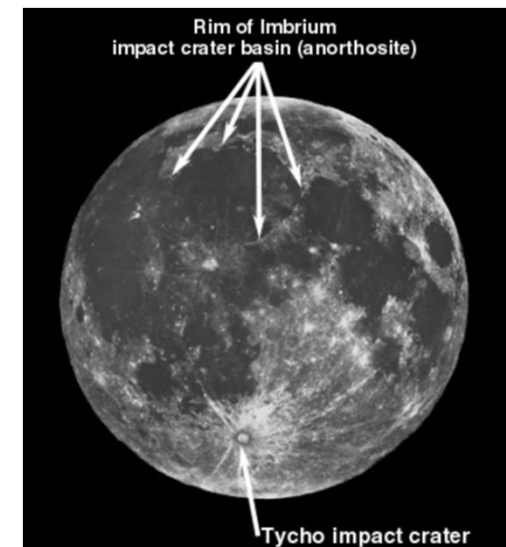


Comets & meteorites likely brought volatiles after the crust had formed

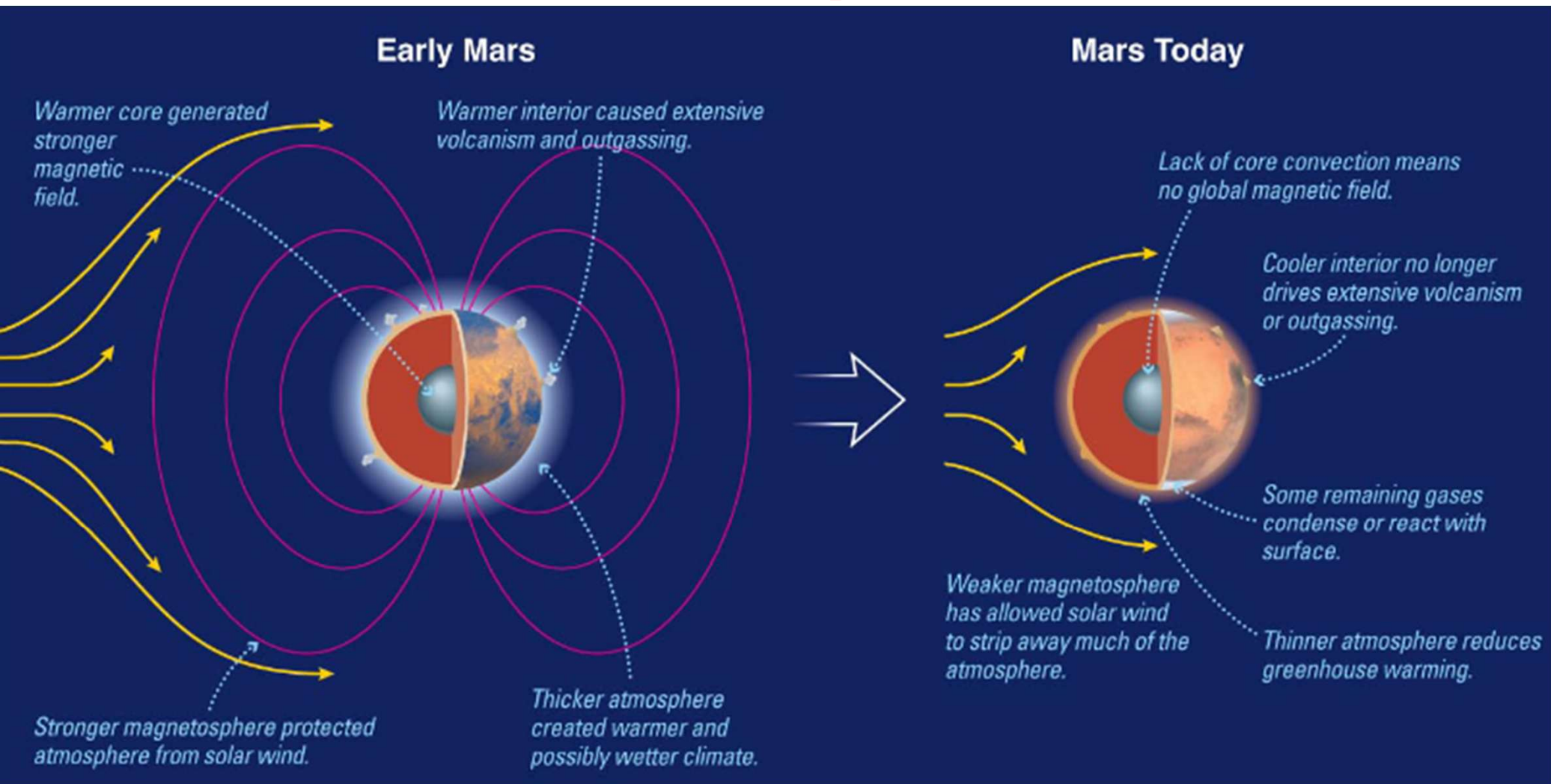
The heavy bombardment



- Early Earth had many collisions with impactors of various sizes. However, due to active plate tectonics most preserved impact craters are “small” and young
- Part of the moon surface is quite old while most of the mare basalts are younger than 3.5 Ga and contain only small craters
- Most huge craters on the Moon are on the older surface and their impact-formed rocks have been dated to 4.1 to 3.8 billion years (Ga) indicating that large impactors were more common in the early Hadean
- It may not be a coincidence that the earliest evidence for life dates into the period when the heavy bombardment slowed down



The atmosphere



- Planets need an magnetic field to hold onto their atmosphere and protect it from solar winds
- The oldest evidence for the Earths magnetic field is 4.2 Ga old.
- Mars had a magentic field 4 Ga ago and lost it around 3.8 Ga ago.

Our Second atmosphere came out of the mantle

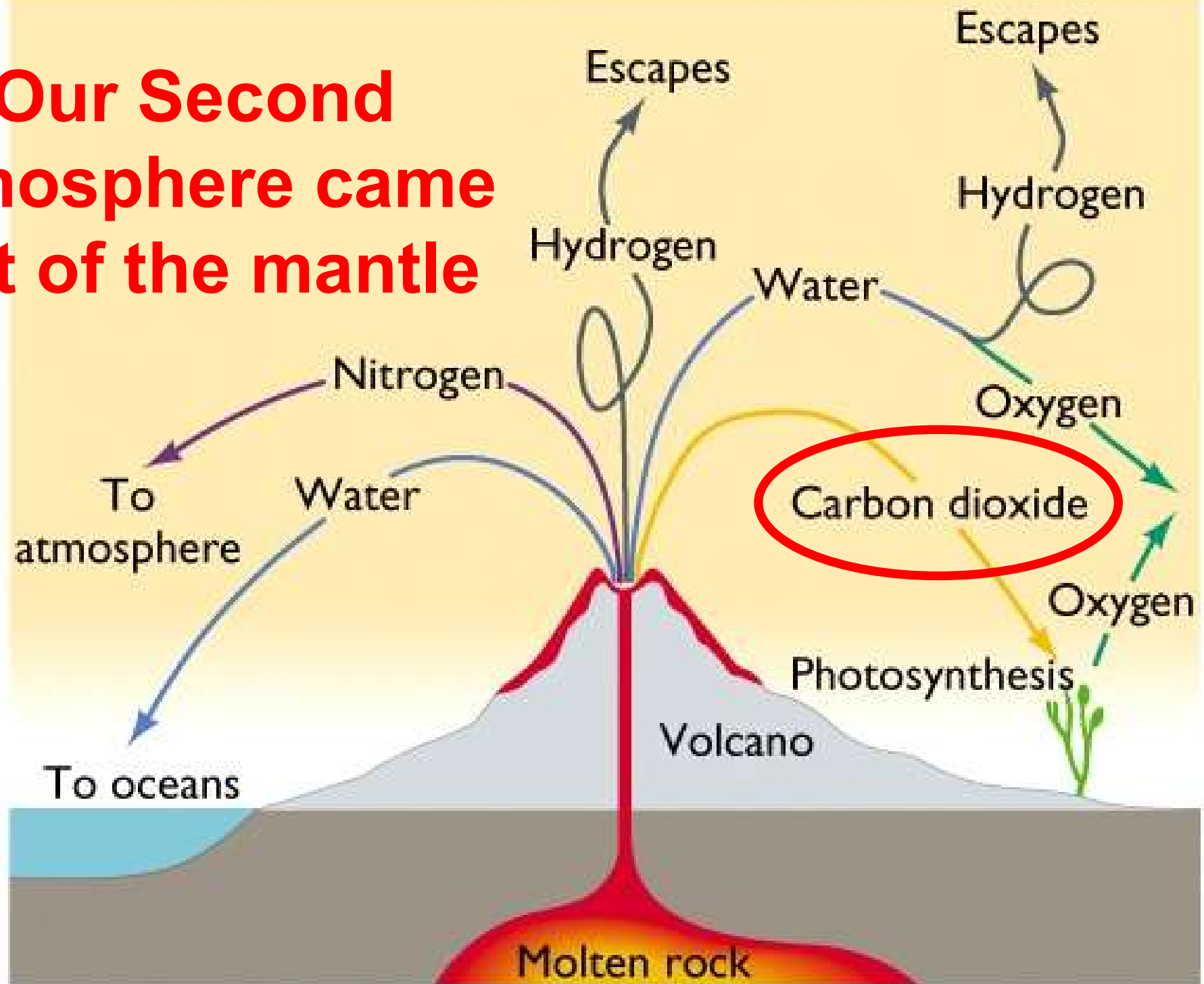


Fig. 1.8

The second atmosphere degassed from mantle magma and comprised primarily of CO₂ with small amounts of organics and water that were delivered by meteorites

	VENUS	EARTH	MARS
SURFACE PRESSURE	100,000 mb	1,000 mb	6 mb
COMPOSITION			
CO ₂	>98%	0.03%	96%
N ₂	1%	78%	2.5%
Ar	1%	1%	1.5%
O ₂	0.0%	21%	2.5%
H ₂ O	0.0%	0.1%	0-0.1%

MARS AND VENUS ARE STUCK WITH THEIR 2ND ATMOSPHERE
THIRD ATMOSPHERE OF EARTH EVOLVED THANKS TO
PHOTOSYNTHETIC BACTERIA THAT USED CO₂ AND PRODUCED
O₂ AND THE MICROBIAL NITROGEN CYCLE

Other planets that teach about early earth

VENUS

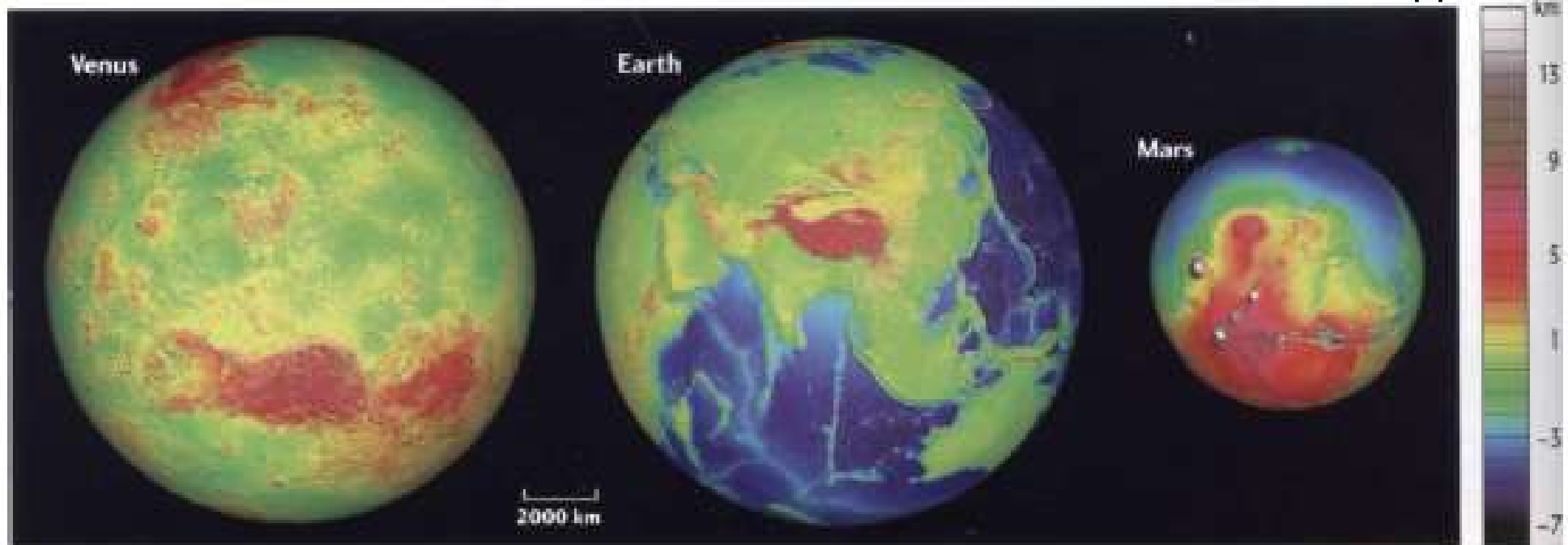
Very hot atmosphere full of GHG. Geologically active with plate tectonics

EARTH

Plate tectonics
Geologically active
Photosynthesis => O_2
rich atmosphere
Nitrogen cycle => N_2

MARS

Used to have an atmosphere & ocean till ~3.5 Ga. Too small to keep a Geodynamo going. Thin atmosphere. Plate tectonics stopped



PRACTICE QUESTIONS

You can submit the answer to these questions by Tuesday 19th August if you want to get them corrected before the Midsem exam

MCQ style

- 1) Which planets are considered the “gassy giants”?
 - a) Mercury, Saturn, Uranus, and Neptune
 - b) Jupiter, Saturn, Uranus, and Neptune
 - c) Mercury, Venus, Earth, and Mars
 - d) Earth, Saturn, Uranus, and Neptune
- 2) What process enabled the formation of our inner rocky planets, including our planet Earth?
 - a) The Big Bang
 - b) Accretion
 - c) Collision
 - d) Supernova
- 3) What are the four most common elements that make up the Earth’s crust?
 - a) Oxygen, hydrogen, helium, iron
 - b) Hydrogen and helium
 - c) Oxygen, silicon, helium, iron
 - d) Oxygen, silicon, aluminum, iron

MCQ style

- 1) **What percentage of the Earth is made up of elements other than Hydrogen and Helium?**
 - a) 45%
 - b) 2%
 - c) 90%
 - d) 85%
- 2) **When did our Earth form, approximately?**
 - a) 2.6 billion years ago
 - b) 4.5 billion years ago
 - c) We don't know when the Earth formed.
 - d) 13.8 billion years ago
- 3) **Which of the following is a byproduct of the Earth's metallic core and protects the biosphere from the Sun's most short-wave radiation?**
 - a) The magnetic field
 - b) The atmosphere
 - c) The ozone layer
 - d) The cloud layer

MCQ style

- 1) The early Earth was so hot that it melted. As it did so, the heaviest elements sank to the center while the lightest ones rose. What did the lightest elements form?
 - a) The crust
 - b) The biosphere
 - c) The core
 - d) The atmosphere
- 2) Which of the following best describes the state of the Earth in its infancy, 4.5 billion years ago?
 - a) A hot and inhospitable place for life
 - b) A freezing cold and inhospitable place for life
 - c) A hot but hospitable place for life
 - d) A freezing cold but hospitable place for life
- 3) Radiation from elements created in nearby supernovae, meteor and asteroid impacts during accretion, and pressure generated from chemical differentiation are all reasons for which of the following?
 - a) Why Earth was suitable for the emergence of life
 - b) Why Earth is a rocky, not gaseous planet
 - c) Why Earth only has one moon
 - d) Why early Earth was so hot

MCQ style

- 1) What crucial astronomical factor massively influenced the history of the biosphere? Choose the best answer.
 - a) The impact of Earth's moon on its climate
 - b) The release of greenhouse gases due to movement of tectonic plates
 - c) The impact of photosynthesizing organisms on our atmosphere
 - d) The relationship between the Earth and the Sun
- 2) How do stars, including our Sun, form?
 - a) From a collision between two massive planetary objects
 - b) From a black hole
 - c) From a collapse of a cloud of matter under the pressure of gravity
 - d) Through accretion
- 3) What was it about Earth that allowed it to have just the right amount of energy for the emergence of life?
 - a) Its distance from the Sun
 - b) Its internal core temperature
 - c) The abundance of surface water
 - d) The presence of radioactive chemicals

Long answer style

1. Why do we believe that terrestrial planets looked like primitive chondrites during accretion?
2. Why and how soon after the formation of the solar system did planets and planetesimals melt and into a metallic core and a rocky mantle?
3. How did earth loose its first H & He atmosphere and when?
4. Where did Earth's second atmosphere come from and what gases was it composed off?
5. When and how did Earth's third atmosphere start to form?

Long answer style

1. How did water form on the Earth's surface and where did it come from? Is there more than one theory? Explain.
2. Why does Earth's sister planet Venus not support life?
3. The Early Earth is thought to have been devoid of oxygen. How did oxygen first appear in the Earth's atmosphere?
4. Why and when did Mars lose most of its atmosphere and water?